

Solution Requirements

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Current version does not require user login. Future versions may support secure user authentication for personalized learning profiles.	Low
2	Authorization Levels	The current system provides a single user role with access to all available features. Future versions may include Educator and Student roles.	Low
3	External Interfaces	The system integrates with the Gemini API for generating personalized guidance and with Hugging Face Transformers / PyTorch for BiLSTM and BERT-based emotion classification.	High
4	Transaction Processing	The system processes the user's free-text problem description, predicts the underlying emotion, generates a tailored response, and logs the interaction to CSV.	High
5	Reporting	The system provides an analytics dashboard visualizing emotion trends and side-by-side comparisons between the BiLSTM and BERT models.	Medium
6	Business Rules	Users must provide a description of their study challenge. The system classifies the emotion, applies keyword enhancement, and generates a matching supportive response.	High
7	Compliance to Laws or Regulations	The application does not collect sensitive personal information. It follows general data privacy principles while storing interaction logs locally.	Medium
8	Other	The system stores interaction history in CSV for continuous learning, trend analysis, and future model improvement.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should classify emotion and generate a response quickly, with fast API round-trips.	Average response time < 3 seconds
2	Scalability	The system should support additional learners and future cloud deployment.	Supports at least 100 concurrent users in future deployment
3	Security & Data Privacy	User inputs, logs, and API interactions should be securely handled.	No sensitive personal data stored; secure API communication with Gemini
4	Reliability & Availability	The application should operate continuously without major failures.	System availability > 95% during testing
5	Usability & Accessibility	The interface should be intuitive, empathetic in tone, and easy to navigate for students.	Users receive emotion-aware guidance within 2 interaction steps
6	Maintainability & Portability	The code should be modular, reusable across models, and easy to deploy across environments.	Modular architecture supporting dual models (BiLSTM/BERT) with documented functions